

COMICCRAFT

AI Comic Story Creator using Gemini Models

PHASE 4: Project Planning Phase

Planning Objective

The project is organized into milestones and activities so that the ComicCraft application can be developed systematically from model selection through local deployment.

Milestone 1 – Model Selection and Architecture

- Research suitable generative AI models for storytelling and image generation.
- Compare model capabilities for creative writing, prompt adherence, speed, image quality, documentation, and support.
- Select Gemini Flash, Gemini Pro, and Stable Diffusion for the required tasks.
- Define the overall application architecture and development environment.

Milestone 2 – Core Functionality Development

- Develop the structured comic outline generation function.
- Develop the story narration and dialogue generation function.
- Develop image generation and image-saving functionality.
- Build the comic layout and PDF export functions.
- Implement FastAPI processing for user input.

Milestone 3 – routes.py Development

Create and connect routes for homepage loading, comic generation, JSON-based generation, export confirmation, and image-generation testing.

Milestone 4 – Frontend Development

- Design the input form and comic preview interface.
- Create responsive layouts using HTML and CSS.
- Create dynamic Jinja2 templates for user input, comic preview, and export confirmation.

Milestone 5 – Deployment and Verification

- Create a virtual environment and install dependencies.
- Configure Gemini and Hugging Face API credentials through environment variables.
- Run the application using Uvicorn.
- Verify the complete local comic-generation workflow.

Planning Outcome

The milestone-based plan provides a clear development path from initial research and architecture to implementation, interface development, testing, and local deployment.

Project documentation prepared for phase-wise submission.